

- 1) The relationship between the coefficients and the error term must be linear.
- 2) The residuals must have a mean zero, be independent, be normally distributed, and have variances that are uniform across the variable.
- 3) Variables in SEM should be continuous, interval level data. This means SEM is not appropriate for censored data.
- 4) No specification error. If necessary variables are omitted or unnecessary variables are included in the model, there will be measurement error and the measurement model will not be accurate. Variables included in the model must have acceptable level of kurtosis.

17.5 SIMILARITIES BETWEEN TRADITIONAL STATISTICAL METHODS AND SEM

SEM is similar to traditional methods like correlation, regression and analysis of variance in many ways. First, both traditional methods and SEM are based on linear statistical models. Second, statistical tests associated with both methods are valid if certain assumptions are met. Traditional methods assume a normal distribution and SEM assumes multivariate normality. Third, neither approach offers a test of causality.

17.6 DIFFERENCES BETWEEN TRADITIONAL METHODS AND SEM

Traditional approaches differ from the SEM approach in several areas. **First**, SEM is a highly flexible and comprehensive methodology. This methodology is appropriate for investigating achievement, economic trends, health issues, family and peer dynamics, self-concept, exercise, self-efficacy, depression, psychotherapy, and other phenomenon.

Second, traditional methods specify a default model whereas SEM requires formal specification of a model to be estimated and tested. SEM offers no default model and places few limitations on what types of relations can be specified. SEM model specification requires researchers to support hypothesis with theory or research and specify relations a priori.

Third, SEM is a multivariate technique incorporating observed (measured) and unobserved variables (latent constructs) while traditional techniques analyze only measured variables. Multiple, related equations are solved simultaneously to determine parameter estimates with SEM methodology.

Fourth, SEM allows researchers to recognize the imperfect nature of their measures. SEM explicitly specifies error while traditional methods assume that measurement occurs without error.

Fifth, traditional analysis provides straightforward significance tests to determine group differences, relationships between variables, or the amount of variance explained. SEM provides no straightforward tests to determine model fit. Instead, the best strategy for evaluating model fit is to examine multiple tests (e.g., chi-square, Comparative Fit Index (CFI), Bentler-Bonett Non-normed Fit Index (NNFI), Root Mean Squared Error of Approximation (RMSEA)).

Sixth, SEM resolves problems of multicollinearity. Multiple measures are required to describe a latent construct (unobserved variable). Multicollinearity cannot occur because unobserved variables represent distinct latent constructs.

Finally, a graphical language provides a convenient and powerful way to present complex relationships in SEM. Model specification involves formulating statements about a set of variables. A diagram, a pictorial representation of a model, is transformed into a set of equations. The set of equations are solved simultaneously to test model fit and estimate parameters.

17.7 CONCEPTS AND TERMINOLOGY USED IN SEM

The key elements of essentially all structural equation models are their parameters (often referred to as model parameters or unknown parameters). Model parameters reflect those aspects of a model that are typically unknown to the researcher, at least at the beginning of the analyses, yet are of potential interest to him or her.

Parameter is a generic term referring to a characteristic of a population, such as mean or variance on a given variable, which is of relevance in a particular study.

A **model** is a statistical statement about the relations among variables.

A **path diagram** is a pictorial representation of a model.

A **measured variable** (MV) is a variable that is directly measured whereas a latent variable (LV) is a construct that is not directly or exactly measured.

A **latent variable** (LV) could be defined as whatever its multiple indicators have in common with each other. LVs defined in this way are equivalent to common factors in factor analysis and can be viewed as being free from error of measurement.

Direct effect is a directional relation between two variables, e.g., independent and dependent variables.

Indirect effect is the effect of an independent variable on a dependent variable through one or more intervening or mediating variables.

Specification is formulating a statement about a set of parameters and stating a model.

Fit indices indicate the degree to which a pattern of fixed and free parameters specified in the model is consistent with the pattern of variances and covariances from a set of observed data. Examples of fit indices are chi-square, CFI, NNFI, and RMSEA.

Components of a general structural equation model are the measurement model and the structural model.

The **measurement model** prescribes latent variables, e.g., confirmatory factor analysis. The **structural model** prescribes relations between latent variables and observed variables that are not indicators of latent variables.

Identification involves the study of conditions to obtain a single, unique solution for each and every free parameter specified in the model from the observed data.

The purpose of *estimation* is to obtain numerical values for the unknown (free) parameters.

Exogenous variable—Independent variables not presumed to be caused by variables in the model.

Endogenous variables— variables presumed to be caused by other variables in the model.

Recursive models assume that all causal effects are represented as unidirectional and no disturbance correlations among endogenous variables with direct effects between them.

Non-recursive models are those with feedback loops Model Specification—Formally stating a model via statements about a set of parameters.

Model Identification—Can a single unique value for each and every free parameter be obtained from the observed data.

Evaluation of Fit—Assessment of the extent to which the overall model fits or provides a reasonable estimate of the observed data.

Model Modification—adjusting a specified and estimated model by freeing or fixing new parameters.

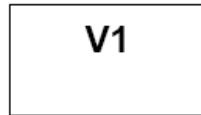
17.7.1 Path Diagrams

One of the easiest ways to communicate a structural equation model is to draw a diagram of it, referred to as *path diagram*, using special graphical notation. A path diagram is a form of graphical representation of a model under consideration. Such a diagram is equivalent to a set of equations defining a model (in addition to distributional and related assumptions), and is typically used as an alternative way of presenting a model pictorially. Path diagrams not only enhance the understanding of structural equation models and their communication among researchers with various backgrounds, but also substantially contribute to the creation of correct command files to fit and test models with specialized programs.

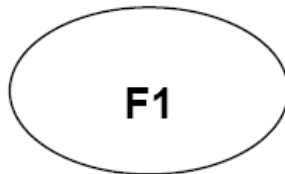
For constructing the path diagrams, widely accepted set of conventions for the graphical display of structural equation models can be used. According to these conventions, unobserved variables, also known as latent variables or constructs are represented as circles or ovals and measured variables also known as observed variables, indicators or manifest variables are represented by squares or rectangles. Lines indicate relations between variables; lack of a line connecting variables implies that no direct relationship has been hypothesized. Lines have either one or two arrows. A line with one arrow represents a hypothesized direct relationship between two variables. The variable with the arrow pointing to it is the DV. A line with an arrow at both ends indicates a covariance between the two variables with no implied direction of effect. Latent variables in SEM generally correspond to hypothetical constructs or factors, which are explanatory variables, supposed to reflect a range that is not directly observable. An example is the construct of intelligence. There is no single, definitive measure of intelligence. Instead, researchers use different types of observed variables, such as tasks of verbal reasoning or memory capacity, to assess various facets of intelligence. Latent variables in SEM can represent a wide range of phenomena. For example, constructs about attributes of people (e.g., intelligence, neuroticism), higher-level units of analysis (e.g.,

groups, geographic regions), or measures, such as method effects (e.g., self-report, observational), can all be represented as latent variables in SEM. An observed variable used as an indirect measure of a construct is referred to as an indicator. The explicit distinction between factors and indicators in SEM allows one to test a wide variety of hypotheses about measurement.

Diagram Symbols



measured variable (V1), observed variable



latent construct (F1), factor, unmeasured variable



direct relationship

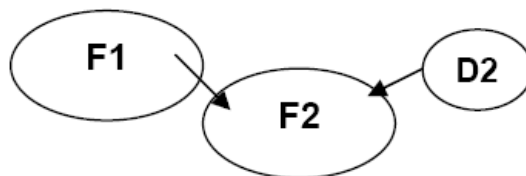
covariance or correlation



error (e1) associated with measured variable (V1)



path coefficient for regression of a latent variable (F1) on an observed variable (V1)



path coefficient for regression of one latent variable (F1) onto another latent variable (F2), residual error (D2) in prediction of F2 by F1.

1.7.2 Classical SEM Notations

These are the c “classical” SEM notation that needs some consideration. ξ (ksi) represents a latent construct associated with observed x_i indicators, η (eta) stands for a latent construct associated with observed y_i indicators, the error terms δ (delta) and ϵ (epsilon) are associated with observed x_i and y_i indicators, respectively. ζ (zeta) is the error term associated with the formative construct. λ_{ij} represents factor loading in the i -th observed indicator that is explained by

the j -th latent construct. γ_{ij} represents weight in the i -th observed indicator that is explained by the j -th latent construct.

Table 17.1: Summary of abbreviations and descriptions used in the SEM study

Symbol	Name	Description
ξ	ksi	A latent construct associated with observed x_i indicators
η	eta	A latent construct associated with observed y_i indicators
δ_i	delta	The error term associated with observed x_i indicators
ε_i	epsilon	The error term associated with observed y_i indicators
ζ	zeta	The error term associated with formative construct
λ_{ij}	lambda	Factor loading in the i -th observed indicator that is explained by the j -th latent construct
γ_{ij}	gamma	Weight in the i -th observed indicator that is explained by the j -th latent construct
x_i		An indicator associated with exogenous construct, i.e. vector of observed exogenous variable
y_i		An indicator associated with endogenous construct, i.e. vector of observed endogenous variable.

17.8 SEM MODELS SPEECIFICATION

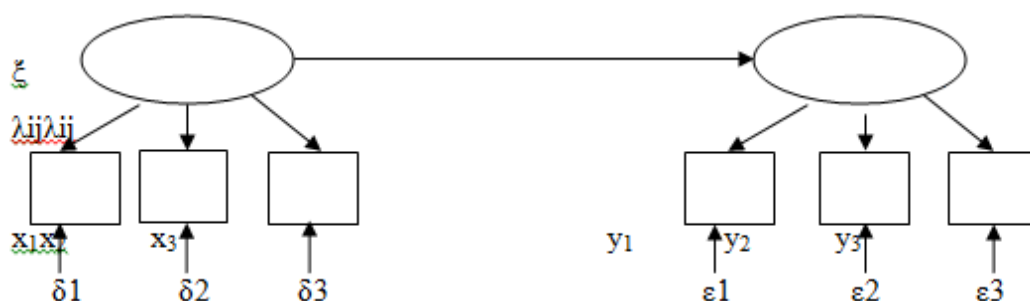
There are three diffeerent SEM models based on different specifications.

17.8.1 Based on Reflective Indicators

The following figure depicts the “classical” SEM case where the model is specified in the reflective mode known as Type A model specification. The case illustrates a path diagram between the two latent constructs (ξ – exogenous and η – endogenous), with three indicators per construct (x_i and y_i). This case can be represented by equations 1 and 2:

- 1) $x_i = \lambda_{ij}\xi + \delta_i$
- 2) $y_i = \lambda_{ij}\eta + \varepsilon_i$

This specification assumes that the error term is unrelated to the latent variable $\text{COV}(\eta, \varepsilon_i) = 0$, and independent $\text{COV}(\varepsilon_i, \varepsilon_j) = 0$, for $i \neq j$ and expected.



Reflective indicators

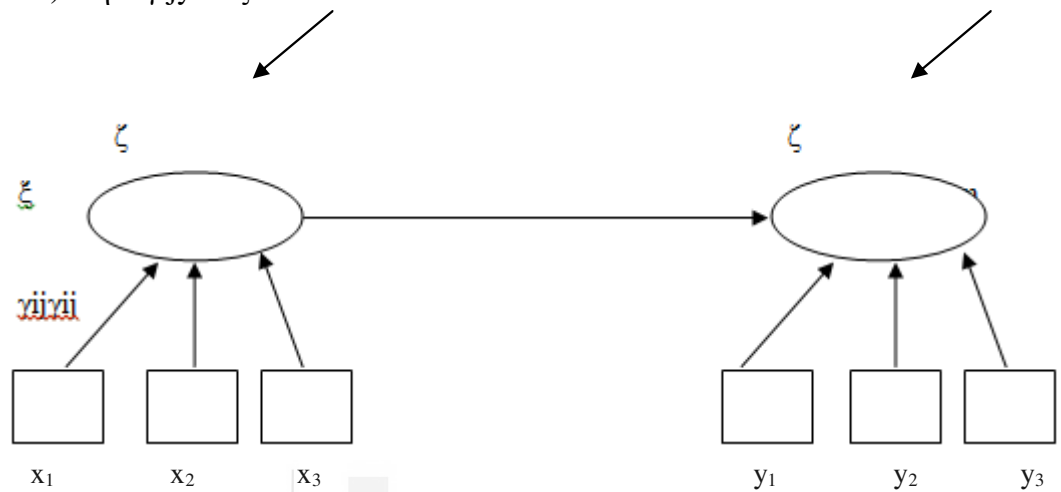
17.8.2 Based on Formative Indicators

Type B model specification, is known as a formative (Fornell&Bookstein 1982; cf. Edwards 2001) or causal indicator (Bollen and Lennox 1991),

because the direction of causality goes from the indicators (measures) to the construct and the error term is estimated at the construct level. This type of model specification can be represented by equations 3 and 4:

$$3) \quad \xi = \gamma_{ij}x_i + \zeta$$

$$4) \quad \eta = \gamma_{ij}y_i + \zeta$$



Formative Indicators

This specification assumes that the indicators and error term are not related, i.e. $\text{COV}(y_i, \zeta) = 0$, and $E(\zeta) = 0$. Formative indicators were introduced for the first time by Curtis and Jackson (1962) and extended by Blalock (1964). This type of model specification assumes that the indicators have an influence on (or that they cause) a latent construct. In other words, the indicators as a group “jointly determine the conceptual and empirical meaning of the construct.”

The type B model specification would give better explanatory power, in comparison to the type A model specification, if the goal is the explanation of unobserved variance in the constructs. A model is identified if model parameters have only one group of values that create the covariance matrix. In order to resolve the problem of indeterminacy that is related to the construct-level error term, the formative-indicator construct must be associated with unrelated reflective constructs. This can be achieved if the formative construct emits paths to:

- i) at least two unrelated reflective indicators;
- ii) at least two unrelated reflective constructs; and
- iii) one reflective indicator that is associated with a formative construct and one reflective construct.

From an empirical point of view, the latent construct captures

- i) the common variance among indicators in the type A model specification; and
- ii) the total variance among its indicators in the type B model specification, covering the whole conceptual domain as an entity (cf. Cenfetelli & Bassellier 2009; MacKenzie et al. 2005).

Reflective indicators are expected to be interchangeable and have a common theme. Interchangeability, in the reflective context, means that omission of an indicator will not alter the meaning of the construct. In other words, reflective measures should be unidimensional and they should represent the common theme of the construct. Formative indicators are not expected to be interchangeable, because each measure describes a different aspect of the construct's common theme and dropping an indicator will influence the essence of the latent

variable. The behavior of measures of the construct with regards to the same antecedents and consequences is an important criterion for the assessment of the construct-indicator relationship. Reflective indicators are interchangeable, which means that measures are affected by the construct, and they must have the same antecedents and consequences. The formative constructs are affected by the measures, thus are not necessarily interchangeable, and each measure can represent a different theme. For the formative indicators, it is not necessary to have the same antecedents and consequences.

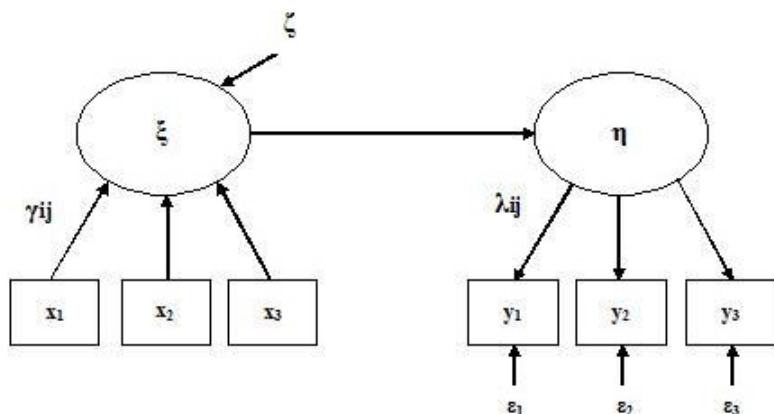
Internal consistency is implied in the reflective indicators because correlation between variables is must. High correlations among the reflective indicators are essential, because they represent the same underlying theoretical concept. This means that all of the items are measuring the same phenomenon within the latent construct. On the contrary, within the formative indicators, internal consistency is not implied because the researcher does not expect high correlations among the measures. Because formative measures are not required to be correlated, validity of construct should not be assessed by internal consistency and reliability as with the reflective measures, but with other means such as nomological and/or criterion-related validity.

17.8.3 Based on both Reflective and Formative Indicators

The researcher can create a model that uses both formative and reflective indicators. It is possible for a structural model to have one type of latent construct at the first-order (latent construct) level and a different type of latent construct at the second-order level (Fornell&Bookstein 1982; MacKenzie et al. 2005; Diamantopoulos & Siguaw 2006; Wetzels et al. 2009). In other words, the researcher can combine different latent constructs to form a hybrid model (Edwards & Bagozzi 2000; McDonald 1996; Tenenhaus et al. 2005). Development of this model type depends on the underlying causality between the constructs and indicators, as well as the nature of the theoretical concept. The researcher should model exogenous constructs in the formative mode and all endogenous constructs in the reflective mode,

- i) if one intends to explain variance in the unobservable constructs (Fornell & Bookstein 1982; cf. Wetzels et al. 2009); and
- ii) in case of weak theoretical background (Wold 1980).

Conducting a VBSEM approach in this model, using a PLS algorithm, is equal to redundancy analysis (Fornell et al. 1988; cf. Chin 1998b), because the mean variance in the endogenous construct is predicted by the linear outputs of the exogenous constructs.



17.9 ISSUES IN SEM TECHNIQUE

17.9.1 Sample Size and Power

SEM is based on covariance. Covariance is less stable when estimated from small samples. Therefore, large sample sizes are needed for SEM analyses. Parameter estimates and chi-square tests of fit are also very sensitive to sample size; therefore, SEM is a large sample technique. However, it may be possible to estimate small models with fewer participants if variables are highly reliable. In addition, although SEM is a large sample technique and test statistics are affected by small samples, promising work has been done by Bentler and Yuan (1999) who developed test statistics for small samples sizes.

17.9.2 Missing Data

Problems of missing data are repeatedly exaggerated in SEM due to the large number of measured variables employed. The researcher who relies on using complete cases only is often left with an inadequate number of complete cases to estimate a model. Therefore missing data attribution is particularly important in many SEM models. When there is evidence that the data are **missing at random** (MAR; missing data on a variable may depend on other variables in the dataset but the missing data does not depend on the variable itself) or **missing completely at random** (MCAR; missing data is unrelated to the variable missing data or the variables in the dataset), the **EM** (expectation maximization) algorithm to obtain **maximum likelihood** (ML) estimates, is appropriate method of imputing missing data. Some of the software packages now include procedures for estimating missing data, including the EM algorithm. EQS 6.1 (Bentler, 2004) produces the EM-based maximum likelihood solution automatically, based on the Jamshidian–Bentler (Jamshidian & Bentler, 1999) computations.

LISREL and AMOS also produce EM-based maximum likelihood estimates. but, if the data are not normally distributed, maximum likelihood test statistics— including those based on the EM algorithm—may be quite inaccurate. Multiple imputation is another option for treatment of missing data.

17.9.2 Multivariate Normality and Outliers

The most commonly employed techniques for estimating models assume multivariate normality in SEM. It is always useful to examine both univariate and multivariate normality indexes to assess normality. Univariate distributions can be examined for outliers and skewness and kurtosis. Multivariate distributions are examined for normality and multivariate outliers. Multivariate normality can be evaluated through the use of Mardia's (1970) coefficient and multivariate outliers can be evaluated through evaluation of Mahalanobis distance. Mardia's (1970) coefficient evaluates multivariate normality through evaluation of multivariate kurtosis. Mardia's coefficient can be converted to a normalized score (equivalent to z score), often normalized coefficients greater than 3.00 are indicative of non-normality. Mahalanobis distance is the distance between a case and the centroid (the multivariate mean with that data point removed). Mahalanobis distance is distributed as a chi-square with degrees of freedom equal to the number of measured variables used to calculate the centroid. Therefore, a multivariate outlier can be defined as a case that is associated with a Mahalanobis distance greater than a critical distance specified typically by a $p < .001$.

Many contemporary treatments introduce SEM not just as a statistical technique but as a process involving several stages:

- 1) Model specification,
- 2) Model estimation,
- 3) Data-model fit assessment, and
- 4) Potential model modification,
- 5) Reporting the Results.

17.10.1 Initial Model Specification

The first step in SEM is model conceptualization or specification. This stage consists of gaining knowledge of the underlying theory on which the particular model will be based. This stage begins with stating the hypothesis in both diagrams and equation form then statistically identifying the model and then evaluating the statistical assumptions that underlie the model.

In this phase the model is specified through a diagram. The relations in the diagram are directly translated into equations and the model is then estimated. One method of model specification is the Bentler–Weeks method (Bentler & Weeks, 1980). In this method every variable in the model, latent or measured, is either an IV or a DV. The parameters to be estimated are the (a) regression coefficients, and (b) the variances and the covariances of the independent variables in the model (Bentler, 2001). In the Bentler–Weeks model only independent variables have variances and covariances as parameters of the model.

A normally tricky and often confusing topic in SEM is identification. In SEM, a model is specified, parameters (variances and covariances of IVs and regression coefficients) for the model are estimated using sample data, and the parameters are used to produce the estimated population covariance matrix. However, only models that are identified can be estimated. A model is said to be identified if there is a unique numerical solution for each of the parameters in the model. The first step consists of counting the number of data points and the number of parameters that we want to estimate. The data in SEM are the variances and covariances in the sample covariance matrix. The number of data points is the number of non redundant sample variances and covariances, where p equals the number of measured variables. The number of parameters is established by adding together the number of regression coefficients, variances, and covariances that are to be estimated. There should be more data points than parameters to be estimated for the estimation of model.

Hypothesized models are said to be over identified if there are more data than parameters to be estimated. The model is said to be just identified if, there are the same numbers of data points as parameters to be estimated. In this case, the estimated parameters perfectly reproduce the sample covariance matrix and the chi-square test statistic and degrees of freedom are equal to zero, hypotheses about the adequacy of the model cannot be tested. However, hypotheses about specific paths in the model can be tested. The model is said to be under identified if there are fewer data points than parameters to be estimated, and parameters cannot be estimated. The number of parameters needs to be reduced

by fixing, constraining, or deleting some of them. A parameter may be fixed by setting it to a specific value or constrained by setting the parameter equal to another parameter. Examining the measurement portion of the model is the second step in determining model identifiability. The measurement part of the model deals with the relationship between the measured indicators and the factors.

Factors, are hypothetical constructs and consist of common variance, as such they have no intrinsic scale and therefore need to be scaled. Measured variables have scales, for example, income may be measured in dollars or weight in pounds. To establish the scale of a factor, the variance for the factor is fixed to 1.00, or the regression coefficient from the factor to one of the measured variables is fixed to 1.00. Fixing the regression coefficient to 1 gives the factor the same variance as the common variance portion of the measured variable. If the factor is an IV, either alternative can be applied. If the factor is a DV the regression coefficient is set to 1.00. Not setting the scale of a latent variable is easily the most common error made when first identifying a model. Next, to establish the identifiability of the measurement portion of the model the number of factors and measured variables are examined. If there is only one factor, the model may be identified if the factor has at least three indicators with nonzero loading and the errors (residuals) are uncorrelated with one another. If there are two or more factors, again consider the number of indicators for each factor.

If each factor has three or more indicators, the model may be identified if errors associated with the indicators are not correlated, each indicator loads on only one factor and the factors are allowed to co-vary. If there are only two indicators for a factor, the model may be identified if there are no correlated errors, each indicator loads on only one factor, and none of the covariances among factors is equal to zero.

17.10.2 Model Estimation

The next step in SEM is to evaluate and estimate population parameters. The purpose of estimation is to minimize the difference between the structured and unstructured estimated population covariance matrices.

$$F = (s - s(Q))W(s - s(Q)),$$

To accomplish this goal a function, F , is minimized where, s is the vector of data (the observed sample covariance matrix stacked into a vector); s is the vector of the estimated population covariance matrix (again, stacked into a vector) and (Q) indicates that s is derived from the parameters (the regression coefficients, variances and covariances) of the model. W is the matrix that weights the squared differences between the sample and estimated population covariance matrix.

In EFA the observed and reproduced correlation matrices are compared. This idea is extended in SEM to include a statistical test of the differences between the estimated structured and unstructured population covariance matrices. If the weight matrix, W , is chosen correctly, at the minimum with the optimal, F multiplied by $(N - 1)$ yields a chi-square test statistic.

There are many different estimation techniques in SEM, these techniques vary by the choice of W .